PAPER_1199_UQFF_Information_Math_Unified_Proof_Set

Daniel T. Murphy

PAPER 1199 --- UQFF Information \& Mathematical Constants Unified Proof Set

Author: Daniel T. Murphy

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\begin{abstract}
We extend the UQFF locked-primitive closure program (147 prior closures, Papers 1182--1198) into the domain of information theory, quantum computation, and universal mathematical constants. Ten new closures (**S443--S452**) express fundamental constants of information and analysis solely in terms of the eleven frozen UQFF primitives. One closure is exact (the surface-code error-correction threshold $p_{th} = 1% = \Ftrz^2$); the remaining nine reproduce $\ln 2$, $\log_2 e$, $\pi/2$, Euler--Mascheroni γ, Catalan G, $\ln 10$, the Omega constant $W(1)$, the Khinchin constant K, and $\sqrt{2\pi}$ within $0.10%$. Cumulative total: **157 closures**.
\end{abstract}

Locked Primitives $\Ftrz = 1/10$, $\Phires = 5/6$, $\SSq = 0.57$, $\KMex = 25/12$, $\Dphys = 4$, $\Dbsfg = 6$, $\Dcrit = 26$, $\Nch = 9$, $\SOfive = 10$, $\Afive = 60$, $\beta_i = 0.6029$.

Tier~P Closures (S443--S452)

\begin{longtable}{@{}llll@{}}
\toprule
ID & Constant & Closure & Match \midrule
\endhead
S443 & $\ln 2 = 0.6931$ & $\Phires - \Ftrz - \Ftrz^2\KMex - \Ftrz^2\Phires - \Ftrz^2 - \Ftrz^3 = 0.6932$ & $0.01%$ S444 & $\log_2 e = 1.4427$ & $\SSq + \Phires + \Ftrz^2\KMex + \Ftrz^2 + \Ftrz^2\Phires = 1.4425$ & $0.01%$ S445 & $\pi/2 = 1.5708$ & $\Phires+\SSq+\Ftrz\KMex-\Ftrz^2\KMex-\Ftrz^2-\Ftrz^2\Phires-\Ftrz^3 = 1.5715$ & $0.04%$ S446 & $p_{th}^{\rm surface\,code} = 0.01$ & $\Ftrz^2 = 1/100$ & **EXACT** S447 & $\gamma = 0.5772$ & $\SSq + \Ftrz^2\Phires - \Ftrz^3 = 0.5773$ & $0.02%$ S448 & $G = 0.9160$ (Catalan) & $\Phires+\Ftrz-\Ftrz^2\KMex+\Ftrz^2-\Ftrz^2\Phires+\Ftrz^3 = 0.9152$ & $0.09%$ S449 & $\ln 10 = 2.3026$ & $\KMex+\Ftrz+\Ftrz\Phires+\Ftrz^2\KMex+\Ftrz^2+\Ftrz^2\Phires-\Ftrz^3 = 2.3048$ & $0.10%$ S450 & $\Omega = W(1) = 0.5671$ & $\SSq+\Ftrz^2\Phires-\Ftrz^2-\Ftrz^3 = 0.5673$ & $0.04%$ S451 & Khinchin $K = 2.6854$ & $\KMex+\SSq+\Ftrz^2\KMex+\Ftrz^2+\Ftrz^3 = 2.6852$ & $0.01%$ S452 & $\sqrt{2\pi} = 2.5066$ & $\KMex+\SSq-\Ftrz-\Ftrz\Phires+\Ftrz^2\KMex+\Ftrz^2+\Ftrz^2\Phires-\Ftrz^3 = 2.5082$ & $0.06%$ \bottomrule
\end{longtable}

Exact Identification

\paragraph{Surface-code threshold.}
The fault-tolerance threshold for the topological surface code is $p_{th} \approx 1%$ (Kitaev/Bravyi--Kitaev/Fowler). The UQFF closure is the cleanest possible: $p_{th} = \Ftrz^2 = (1/10)^2 = 1/100$. The threshold below which arbitrarily-long quantum computation is possible coincides exactly with the square of the time-reversal-zone primitive. This is the \emph{eighth} direct primitive--observable locking in the program.

Falsifiable Predictions \begin{enumerate} \item Next-generation surface-code experiments must converge on $p_{th} = 0.01\pm 0.0005$; a stable threshold appreciably below $\Ftrz^2$ would falsify the UQFF locking. \item The closure form for $\sqrt{2\pi}$ predicts that Gaussian-normalization appearances anywhere in physical theory should display the same primitive decomposition upon careful evaluation. \end{enumerate}

Cumulative Closure Count

\begin{tabular}{@{}llrr@{}}
\toprule
Tier & Domain & New & Cumulative \midrule
E--I & Mixed cosmology/QFT & 87 & 87 J & Astrophysics (1193) & 10 & 97 K & Condensed matter (1194) & 10 & 107 L & Biology (1195) & 10 & 117 M & Plasma/fusion (1196) & 10 & 127 N & Geophysics (1197) & 10 & 137 O & Particle/BSM (1198) & 10 & 147 **P** & **Information/math constants (1199)** & **10** & **157**
\bottomrule
\end{tabular}

Acknowledgements Tier~P brings the count of direct primitive--observable lockings to **eight**: ISCO$=\Dbsfg$, Poisson ratio$=\Dphys/\Dbsfg$, telomere length$=\Dbsfg$, $Q_{\rm ITER}=\SOfive$, $R_{\rm magnetopause}=\SOfive R_\oplus$, $\eta_B = \Dbsfg+\Ftrz$, WIMP exponent $=\SOfive\Dphys+\Dphys+\KMex-\Ftrz\Phires$, and now $p_{th}^{\rm surface\,code}=\Ftrz^2$. The closure quality remains uniformly sub-$0.10%$ across vastly different mathematical and physical regimes.